

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1402
Design

COURSE NAME: Compiler

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

CLASS TEST

Code of Conduct:

I KAVINMITHA .S (192511285) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kavinmitha.s

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y *z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Construct three address code using the following production for Boolean expression (a > b) or (c < d) or (e < f)

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \ \&\& \ B2$

$B \rightarrow ! B1$

$B \rightarrow E1 \text{ relop } E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Q4. Generate the three address code for the following program fragment-

while (A < C and B > D) do

if A = 1 then C = C + 1

else while A < = D do

A = A + B

18/06/20

CO4 AT 1 - CLASS TEST

Name : KAVINMITHA.S

Course code : CSA1402

Reg No : 192511285

Course name : Compiler design.

1. Construct Three address code :

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

38/50

Ans: Given:

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

conditions :

$a < b$

$c < d$

$a > d$

Labels : L_1, L_2, L_3, L_4, L_5

L_1 :

if $a < b$ goto L_2

goto L_5

L2:

if $c < d$ goto L4

goto L3

L3:

if $(a > d)$ goto L4

goto L5

Three address code

L4 $\Rightarrow z = x + y * z$

$\therefore$ L4:

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

goto L6

L5 $\Rightarrow z = z + 1$

$\therefore$ L5:

$t_3 = z + 1$

$z = t_3$

2) Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using Back patching.

Ans:

$[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

if $(a < b)$ goto 3

goto 5

if $(c < d)$ goto 7

goto 5

if $(e < f)$ goto 7

goto 8

7. True

8. False

3) Construct three address code using the following production for boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$.

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ rel op } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

$L_1:$

if $(a > b)$ goto L_{True}
goto L_2

$L_2:$

if $(c < d)$ goto L_{True}
goto L_4

$L_3:$

if $e < f$ goto L_{True}
goto L_{False}

$L_{\text{True}}:$

// Boolean expression is true

$L_{\text{False}}:$

// Boolean expression is False.

4. Generate the three address code for following:

while $(A < C \text{ and } B > D)$ do

if $A = 1$ then $C = C + 1$

else while $A \leq D$ do

$A = A + B$

L1:

if $A < C$ goto L2

goto LEND

L2:

if $B > D$ goto L3

goto LEND

L3:

if $A == 1$ goto L4

goto L5

L4:

$t_1 = C + 1$

$C = t_1$

goto L1

L5:

if $A \leq D$ goto L6

goto L1

L6:

$t_2 = A + B$

$A = t_2$

goto L5

LEND:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps